

MKGW3 BLE to POE/WiFi Gateway

MQTT Protocol

Version V2.3

Contents

1. About This Manual 1
2. MQTT Commands Format 1
    2.1 Setup gateway parameters 1
    2.2 Get gateway parameters 2
    3.3 Gateway report data to cloud 2
    3.2 Result Code Description 2
3. Command List 3
    3.1 System parameters 3
        3.3.1 Reboot device (000) 3
        3.3.2 Device information (002) 4
        3.3.3 Network status report interval (003) 5
        3.3.4 Report network status (004) 6
        3.3.5 Upgrade WIFI Firmware (006) 7
        3.3.6 WIFI firmware upgrade result (007) 7
        3.3.7 NTP setting (008) 8
        3.3.8 System time (009) 9
        3.3.9 LED setting (011) 10
        3.3.10 Firmware and certificate update status (012) 11
        3.3.11 Factory reset (013) 12
        3.3.12 Reset Notification (014) 13
        3.3.13 Upgrade Bluetooth firmware (015) 13
        3.3.14 Bluetooth firmware upgrade result (016) 14
    3.2 Network settings 15
        3.2.1 WIFI parameters (020) 15
        3.2.2 EAP certificates (021) 16
        3.2.3 EAP certificate download result (022) 17
        3.2.4 WIFI IP address (023) 18
        3.2.5 Network type (024) 19
        3.2.6 Ethernet IP address (025) 20
    3.3 MQTT settings 22
        3.3.1 MQTT parameters (030) 22
        3.3.2 MQTT certificates (031) 24
        3.3.3 SSL certificate download result (032) 25
    3.4 Bluetooth parameters 25
        3.4.1 Bluetooth scan switch (040) 25
        3.4.2 Filter relation (041) 26
        3.4.3 RSSI filter (042) 28
        3.4.4 Mac address filter (043) 29

3.4.5 ADV name filter (044) 30
3.4.6 Raw data filter switch (045) 32
3.4.7 iBeacon filter (046) 34
3.4.8 Eddystone-UID filter (047) 35
3.4.9 Eddystone-URL filter (048) 37
3.4.10 Eddystone-TLM filter (049) 38
3.4.11 BXP- device info filter (050) 39
3.4.12 BXP-ACC filter (051) 40
3.4.13 BXP-TH filter (052) 41
3.4.14 BXP-Button filter (053) 42
3.4.15 BXP-Tag filter (054) 44
3.4.16 MK PIR filter (055) 45
3.4.17 Other type filter (056) 47
3.4.18 Duplicate data filter (057) 49
3.4.19 Bluetooth data report setting (059) 51
3.4.20 PHY filter (060) 52
3.4.21 Gateway advertisement settings(061) 53
3.4.22 MK TOF Filter (062) 55
3.4.23 Bluetooth data report interval (063) 56
3.4.24 NanoBeacon filter (064) 57
3.4.25 Data parsing Setting (065) 59
3.4.26 Bluetooth Scan Mode(066) 61
3.5 Report Bluetooth data (070) 63
3.5.1 iBeacon frame 64
3.5.2 Eddystone-UID frame 64
3.5.3 Eddystone-URL frame 64
3.5.4 Eddystone-TLM frame 65
3.5.5 BXP-device info frame 65
3.5.6 BXP-ACC frame 66
3.5.7 BXP-TH frame 66
3.5.8 BXP-Button frame 67
3.5.9 BXP-Tag frame 67
3.5.10 MK PIR frame 68
3.5.11 MK TOF frame 69
3.5.12 NanoBeacon info frame 69
3.5.13 Other type frames 70
4. Revision History 70

1. About This Manual

This document describes the communication protocol between the gateway and the cloud. The communication between the gateway and the cloud is bidirectional, supporting both remote management of the gateway from the cloud and data transmission from the gateway to the cloud.

2. MQTT Commands Format

The command includes a message id which is used to distinguish different actions. Message IDs starting from 1000 are used for gateway parameter configuration, starting from 2000 are for gateway parameter queries, starting from 3000 are for the gateway actively report data to the cloud.

2.1 Setup gateway parameters

Send from cloud

```
{
      "msg_id": 1XXX,       //Message id, starts from 1000 in setup command
     "device_info":{
          "mac":"112233445566"         //The WIFI MAC address of the gateway
     },
       "data":{
              "XX" : XX              //Parameter name and value
       }
}
```

Gateway reply

```
{
      "msg_id":1XXX,              //Message id
     "device_info":{
          "mac":"112233445566"          //The WIFI MAC address of the gateway
     },
     "result_code":X,               // Return code
     "result_msg":"XX"              // Description of the code
}
```

2.2 Get gateway parameters

Send from cloud

```
{
      "msg_id": 2XXX,       //Message id, starts from 2000 in inquire command
     "device_info":{
          "mac":"112233445566"         //The WIFI MAC address of the gateway
       }
}
```

Gateway reply

```
{
      "msg_id":2XXX,              //Message id
     "device_info":{
          "mac":"112233445566"          //The WIFI MAC address of the gateway
     },
      “data”:{
           "XX":XX,                 // Parameter name and value
      }
}
```

3.3 Gateway report data to cloud

Gateway will actively report data to cloud in some situations. For example, reports BLE scan data, WiFi scan data, reports OTA result and so on.

The reporting data format is as below:

```
{
      "msg_id":3XXX,    //Message id, starts from 3000 in report command
     "device_info":{
          "mac":"112233445566"      //The WIFI MAC address of the gateway
     },
       "data":{
               "XX" : XX                 //Data and value
       }
}
```

3.2 Result Code Description

When a command is sent to the gateway, the gateway will reply to each command. There is a return code included in the reply to tell users the setup result. Following are the result code and their descriptions:

result_code	result_msg	description
0	success	Receive and parse the message successfully
1	length error	The data length is out of range
2	type error	The data type doesn't conform the rule
3	range error	The data value is out of range
4	no object error	The parameter name is not found

3. Command List

In the setup command, the message id is 1XXX. In the inquire command, the message id is 2XXX. In the notify command, the message id is 3XXX, the XXX is defined as the message id offset.

3.1 System parameters

3.3.1 Reboot device (000)

This command is used to reboot the gateway.

Flag	Msg id offset	Parameter name	Parameter value
Write	0	reset	number. 0: reboot

Setup:

```
{
    "msg_id": 1000,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
    "reset": 0
}
}
```

Device reply:

```
{
    "msg_id": 1000,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "result_code": 0,
```

```
    "result_msg": "success"
}
```

3.3.2 Device information (002)

To read device information of the gateway.

Flag	Msg id offset	Parameter name	Parameter value
Read	002	ble_mac	String. Bluetooth mac address
		eth_mac	String. Ethernet mac address
		device_name	String.
		product_model	String.
		company_name	String.
		hardware_version	String
		software_version	String. Main controller SDK version
		function_version	**String. Added in V2.X.X firmware**
		firmware_version	String. WiFi firmware version
		sl_ble_version	String. Bluetooth firmware version

Inquire:

```
{
    "msg_id": 2002,
    "device_info": {
        "mac": "4c11ae8be624"
  }
}
```

Device reply:

```
{
  "msg_id" : 2002,
  "device_info" : {
    "mac" : "fcb4675321bc"
  },
  "data" : {
    "ble_mac" : "84fce61492fd",
    "eth_mac" : "fcb4675321bf",
    "device_name" : "MKGW3-92FD",
    "product_model" : "MKGW3",
    "company_name" : "MOKO TECHNOLOGY LTD.",
    "hardware_version" : "V1.0",
    "software_version" : "V4.4.4",
```

```
    "function_version"  :  "V2.0",
    "firmware_version"  :  "V1.0.1",
    "sl_ble_version"  :  "V1.0.3.3"
  }
}
```

3.3.3 Network status report interval (003)

After the gateway is connected to cloud, it will actively report network status to cloud. This command is used to define the time interval of gateway reporting network status to cloud.
When set the interval to 0, the gateway will report network status only once when it is connected to cloud.

Flag	Msg id offset	Parameter name	Parameter value
Write/read	3	report_interval	number. Range: 0 or 30-86400, second Default: 30

Inquire:

```
{
    "msg_id":  2003,
    "device_info":  {
        "mac":  "4c11ae8be624"
     }
}
```

Device reply

```
{
    "msg_id":  2003,
    "device_info":  {
        "mac":  "4c11ae8be624"
    },
    "data":  {
        "report_interval":  30
    }
}
```

Setup:

```
{
    "msg_id":  1003,
    "device_info":  {
        "mac":  "4c11ae8be624"
    },
```

```
    "data": {
        "report_interval": 30
    }
}
```

Device reply:

```
{
    "msg_id": 1003,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "result_code": 0,
    "result_msg": "success"
}
```

3.3.4 Report network status (004)

When the network status report interval comes, the gateway will report current network status to cloud.

Flag	Msg id offset	Parameter name	Parameter value
Notify	4	timestamp	number. Unix timestamp
		timezone	number. -24~28, step by half timezone.
		eth_mac	string
		ble_mac	string
		net_interface	number. 0-1 0: ethernet 1: wifi
		wifi_rssi	number. Unit is dBm Valid when the net_interface is wifi

Device notify:

```
{
    "msg_id": 3004,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "timestamp": 1689219704,
        "timezone": 0,
        "eth_mac": AABBCCDDEEFF,
        "ble_mac": 4c11ae8be624,
```

```
            "net_interface": 1,
            "wifi_rssi": -51
        }
}
```

3.3.5 Upgrade WIFI Firmware (006)

This command is used to upgrade the firmware of WIFI module, the gateway will download firmware upgrade file from the HTTP(s) server for upgrading.

Flag	Msg id offset	Parameter name	Parameter value
Write	6	firmware_url	String, 1-256 characters. http and https are both supported.

Setup:

```
{
    "msg_id": 1006,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "firmware_url": "http://47.104.172.255:8080/esp32.bin"
    }
}
```

Device reply:

```
{
    "msg_id": 1006,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "result_code": 0,
}
```

3.3.6 WIFI firmware upgrade result (007)

The gateway will actively report the upgrade result after WIFI firmware upgrade is completed.

Flag	Msg id offset	Parameter name	Parameter value
Notify	7	result_code	number. -1~2 -1: under updating 0: OTA failed 1: OTA succeed

			2: Download firmware failed (Added in V2.X.X firmware)

```
{
    "msg_id": 3007,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "result_code": 1
    }
}
```

3.3.7 NTP setting (008)

To inquire and setup the NTP function switch and NTP server for the gateway.

Flag	Msg id offset	Parameter name	Parameter value
Write/read	8	switch_value	number. 0-1 0: off 1: on default: 1
		server	string 0-64 characters

Inquire:

```
{
    "msg_id": 2008,
    "device_info": {
        "mac": "4c11ae8be624"
    }
}
```

Device reply:

```
{
    "msg_id": 2008,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "switch_value": 1,
        "server": "www.baidu.com"
            }
}
```

Setup:

```
{
    "msg_id": 1008,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "switch_value": 1,
        "server": "www.baidu.com"
    }
}
```

Device reply:

```
{
    "msg_id": 1008,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "result_code": 0,
    "result_msg": "success"
}
```

3.3.8 System time (009)

To inquire and setup the timestamp and timezone of the gateway.

Flag	Msg id offset	Parameter name	Parameter value
Write/read	9	timestamp	number. Unix timestamp
		timezone	number. -24~28, step by half timezone

Inquire:

```
{
    "msg_id": 2009,
    "device_info": {
        "mac": "4c11ae8be624"
    }
}
```

Device reply:

```
{
    "msg_id": 2009,
    "device_info": {
```

```
        "mac": "4c11ae8be624"
    },
    "data": {
        "timestamp": 1681,
        "timezone": 0
    }
}
```

Setup:

```
{
    "msg_id": 1009,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "timestamp": 1681,
        "timezone": 0
    }
}
```

Device reply:

```
{
    "msg_id": 1009,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "result_code": 0,
    "result_msg": "success"
    }
```

3.3.9 LED setting (011)

To inquire and setup the LED switch of the gateway.

Flag	Msg id offset	Parameter name	Parameter value
Write/read	11	net_led	number. 0-1 0: off, 1: on
		sys_led	number. 0-1 0: off, 1: on
		server_led	number. 0-1 0: off, 1: on

Inquire:

```
{
    "msg_id": 2011,
```

```
    "device_info": {
        "mac": "4c11ae8be624"
    }
}
```


Device reply:


```
{
    "msg_id": 2011,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "net_led": 1,
        "sys_led": 1,
        "server_led": 1
    }
}
```


Setup:


```
{
    "msg_id": 1011,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "net_led": 1,
        "sys_led": 1,
        "server_led": 1
    }
}
```


Device reply:


```
{
    "msg_id": 1011,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "result_code": 0,
    "result_msg": "success"
}
```


3.3.10 Firmware and certificate update status (012)

To inquire firmware upgrade status or certificate download status.

Flag	Msg id offset	Parameter name	Parameter value

Read	12	status	number.0-1 0: not upgrade or download 1: under upgrading or downloading

Inquire:

```
{
    "msg_id": 2012,
    "device_info": {
        "mac": "4c11ae8be624"
    }
}
```

Device reply:

```
{
    "msg_id": 2012,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "status": 0
    }
}
```

3.3.11 Factory reset (013)

To reset the gateway. After reset, the gateway parameters will be restored to factory settings.

Flag	Msg id offset	Parameter name	Parameter value
Write	13	factory_reset	number. 0: reset

Setup:

```
{
    "msg_id": 1013,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "factory_reset": 0
    }
}
```

Device reply:

```
{
```

```
    "msg_id": 1013,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "result_code": 0,
    "result_msg": "success"
}
```

3.3.12 Reset Notification (014)

When the gateway is reset by the reset button, it will actively report a notification before it runs factory reset.

Flag	Msg id offset	Parameter name	Parameter value
Notify	14	/	/

```
{
    "msg_id": 3014,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
    }
}
```

3.3.13 Upgrade Bluetooth firmware (015)

This command is used to upgrade the firmware of Bluetooth module, the gateway will download firmware upgrade file from the HTTP(s) server for upgrading.

Flag	Msg id offset	Parameter name	Parameter value
Write	15	firmware_url	String, 1-256 characters. http and https are both supported.

Setup:

```
{
    "msg_id": 1015,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "firmware_url": "http://47.104.172.255:8080/Bluetooth.bin"
    }
```

```
}
```

Device reply:

```
{
    "msg_id": 1015,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "result_code": 0,
}
```

3.3.14 Bluetooth firmware upgrade result (016)

The gateway will actively report the upgrade result after the Bluetooth firmware upgrade is completed.

Flag	Msg id offset	Parameter name	Parameter value
Notify	16	result_code	number. -1~2 -1: under updating 0: OTA failed 1: OTA succeed **2: Download firmware failed (Added in V2.X.X firmware)**

Device notify:

```
{
    "msg_id": 3016,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "result_code": 1
    }
}
```

3.2 Network settings

To read and change the network settings for the gateway. After sending these commands, and then send the reboot command (msg id: 1000), the gateway will reboot and apply the new WIFI to reconnect.

3.2.1 WIFI parameters (020)

Flag	Msg id offset	Parameter name	Parameter value
Write/read	20	security_type	number. 0-1 0: personal 1: enterprise
		ssid	string 1-32 characters
		passwd	string 0-64 characters
		eap_type	number. 0-2 0: PEAP-MSCHAPV2 1: TTLS-MSCHAPV2 2: TLS
		eap_id	string 0-64 characters
		eap_username	string 0-32 characters
		eap_passwd	string 0-64 characters
		eap_verify_server	number. 0-1 Valid when eap type is 0 0r 1

Inquire:

```
{
    "msg_id": 2020,
    "device_info": {
        "mac": "4c11ae8be624"
    }
}
```

Device reply:

```
{
    "msg_id": 2020,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
```

```
            "security_type": 0,
            "ssid": "moko1234",
            "passwd": "12345678",
            "eap_type": 0,
            "eap_id": "",
            "eap_username": "",
            "eap_passwd": "",
            "eap_verify_server": 0
        }
}
```

Setup:

```
{
    "msg_id": 1020,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "security_type": 0,
        "ssid": "moko1234",
        "passwd": "12345678",
        "eap_type": 0,
        "eap_id": "",
        "eap_username": "",
        "eap_passwd": "",
        "eap_verify_server": 0
    }
}
```

Device reply:

```
{
    "msg_id": 1020,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "result_code": 0,
    "result_msg": "success"
}
```

3.2.2 EAP certificates (021)

To change the EAP certificates, valid when the WIFI security is enterprise.

Flag	Msg id offset	Parameter name	Parameter value

Write	21	ca_url	string. Each url is 0-256 characters. These three parameters can not be all empty.
		client_cert_url	
		client_key_url	

Setup:

```
{
    "msg_id": 1021,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "ca_url": "http://47.104.172.255:8080/ca.pem",
        "client_cert_url": "http://47.104.172.255:8080/client_cert.pem",
        "client_key_url": "http://47.104.172.255:8080/client_key.key"
    }
}
```

Device reply:

```
{
    "msg_id": 1021,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "result_code": 0,
    "result_msg": "success"
}
```

3.2.3 EAP certificate download result (022)

To gateway will actively notify the result after it downloads the EAP certificates.

Flag	Msg id offset	Parameter name	Parameter value
Notify	22	result_code	number. -1: under process 0: failed 1: success

Device notify:

```
{
    "msg_id": 3022,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
```

```
        "result_code":  1
    }
}
```

3.2.4 WIFI IP address (023)

To inquire and change the ip allocation and ip address for WIFI connection.

Flag	Msg id offset	Parameter name	Parameter value
Write/read	23	dhcp_en	number. 0-1 0: disable 1: enable
		ip	string. 7-15 characters
		netmask	string. 7-15 characters
		gw	string. 7-15 characters
		dns	string. 7-15 characters

Inquire:

```
{
    "msg_id":  2023,
    "device_info":  {
        "mac":  "4c11ae8be624"
    }
}
```

Device reply:

```
{
    "msg_id":  2023,
    "device_info":  {
        "mac":  "4c11ae8be624"
    },
    "data":  {
        "dhcp_en":  1,
        "ip":  "0.0.0.0",
        "netmask":  "0.0.0.0",
        "gw":  "0.0.0.0",
        "dns":  "0.0.0.0"
    }
}
```

Setup:

```
{
    "msg_id":  1023,
    "device_info":  {
```

```
        "mac":  "4c11ae8be624"
    },
    "data":  {
        "dhcp_en":  1,
        "ip":  "192.169.0.10",
        "netmask":  "255.255.255.0",
        "gw":  "192.168.0.1",
        "dns":  "8.8.8.8"
    }
}
```

Device reply:

```
{
    "msg_id":  1023,
    "device_info":  {
        "mac":  "4c11ae8be624"
    },
    "result_code":  0,
    "result_msg":  "success"
}
```

3.2.5 Network type (024)

To inquire and change the network connection type.

Flag	Msg id offset	Parameter name	Parameter value
Write/read	24	net_interface	number. 0-3 0: ethernet 1: wifi 2: ethernet->wifi 3: wifi->ethernet

Inquire:

```
{
    "msg_id":  2024,
    "device_info":  {
        "mac":  "4c11ae8be624"
    }
}
```

Device reply:

```
{
    "msg_id":  2024,
    "device_info":  {
        "mac":  "4c11ae8be624"
    },
```

```
    "data": {
        "net_interface": 1
    }
}
```

Setup:

```
{
    "msg_id": 1024,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "net_interface": 1
    }
}
```

Device reply:

```
{
    "msg_id": 1024,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "result_code": 0,
    "result_msg": "success"
}
```

3.2.6 Ethernet IP address (025)

To inquire and change the ip allocation and ip address for Ethernet connection.

Flag	Msg id offset	Parameter name	Parameter value
Write/read	25	dhcp_en	number. 0-1 0: disable 1: enable
		ip	string. 7-15 characters
		netmask	string. 7-15 characters
		gw	string. 7-15 characters
		dns	string. 7-15 characters

Inquire:

```
{
    "msg_id": 2025,
    "device_info": {
        "mac": "4c11ae8be624"
```

```
    }
}
```

Device reply:

```
{
    "msg_id":  2025,
    "device_info":  {
        "mac":  "4c11ae8be624"
    },
    "data":  {
        "dhcp_en":  1,
        "ip":  "0.0.0.0",
        "netmask":  "0.0.0.0",
        "gw":  "0.0.0.0",
        "dns":  "0.0.0.0"
    }
}
```

Setup:

```
{
    "msg_id":  1025,
    "device_info":  {
        "mac":  "4c11ae8be624"
    },
    "data":  {
        "dhcp_en":  1,
        "ip":  "192.169.0.10",
        "netmask":  "255.255.255.0",
        "gw":  "192.168.0.1",
        "dns":  "8.8.8.8"
    }
}
```

Device reply:

```
{
    "msg_id":  1025,
    "device_info":  {
        "mac":  "4c11ae8be624"
    },
    "result_code":  0,
    "result_msg":  "success"
}
```

3.3 MQTT settings

To inquire and change the MQTT settings for the gateway. After sending these commands, and then send the reboot command (**msg id: 1000**), the gateway will reboot and apply the new MQTT settings to reconnect.

3.3.1 MQTT parameters (030)

Flag	Msg id offset	Parameter name	Parameter value
Write/read	30	security_type	number. 0-3 0: no encryption 1: CA signed server certificate 2: CA certificate 3: Self signed certificates
		host	string. 1-64 characters
		port	number. 1-65535
		client_id	string. 1-64 characters
		username	string. 0-256 characters
		passwd	string. 0-256 characters
		sub_topic	string. 1-128 characters
		pub_topic	string. 1-128 characters
		qos	number. 0-2
		clean_session	number. 0-1
		keepalive	number. 10-120, second
		lwt_en	number. 0-1
		lwt_qos	number. 0-2
		lwt_retain	number. 0-1
		lwt_topic	string. 1-128 characters
		lwt_payload	string. 1-128 characters

Inquire:

```
{
    "msg_id": 2030,
    "device_info": {
        "mac": "4c11ae8be624"
    }
}
```

Device reply:


```
{
    "msg_id": 2030,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "security_type": 0,
        "host": "47.11.11.11",
        "port": 1883,
        "client_id": "4c11ae8be624",
        "username": "",
        "passwd": "",
        "sub_topic": "/MK110/4c11ae8be624/receive",
        "pub_topic": "/MK110/4c11ae8be624/send",
        "qos": 0,
        "clean_session": 1,
        "keepalive": 60,
        "lwt_en": 0,
        "lwt_qos": 1,
        "lwt_retain": 0,
        "lwt_topic": "/MK110/4c11ae8be624/send",
        "lwt_payload": "offilne"
    }
}
```


Setup:


```
{
    "msg_id": 1030,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "security_type": 0,
        "host": "47.11.11.11",
        "port": 1883,
        "client_id": "4c11ae8be624",
        "username": "",
        "passwd": "",
        "sub_topic": "/MK110/4c11ae8be624/receive",
        "pub_topic": "/MK110/4c11ae8be624/send",
        "qos": 0,
        "clean_session": 1,
        "keepalive": 60,
```

```
            "lwt_en": 0,
            "lwt_qos": 1,
            "lwt_retain": 0,
            "lwt_topic": "/MK110/4c11ae8be624/send",
            "lwt_payload": "offilne"
        }
}
```

Device reply

```
{
    "msg_id": 1030,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "result_code": 0,
    "result_msg": "success"
}
```

3.3.2 MQTT certificates (031)

To change the SSL certificates, valid when the SSL/TLS is enabled.

Flag	Msg id offset	Parameter name	Parameter value
Write	31	ca_url	string. Each url is 0-256 characters. The three parameters can not be all empty.
		client_cert_url	
		client_key_url	

Setup:

```
{
    "msg_id": 1031,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "ca_url": "http://47.104.172.255:8080/ca.pem",
        "client_cert_url": "http://47.104.172.255:8080/client_cert.pem",
        "client_key_url": "http://47.104.172.255:8080/client_key.key"
    }
}
```

Device reply

```
{
    "msg_id": 1031,
    "device_info": {
        "mac": "4c11ae8be624"
```

```
    },
    "result_code":  0,
    "result_msg":  "success"
}
```

3.3.3 SSL certificate download result (032)

The gateway will actively report the result after it downloads the SSL certificates.

Flag	Msg id offset	Parameter name	Parameter value
Notify	32	result_code	number. -1: under process 0: failed 1: success

Device notify:

```
{
    "msg_id":  3032,
    "device_info":  {
        "mac":  "4c11ae8be624"
    },
    "data":  {
        "result_code":  1
    }
}
```

3.4 Bluetooth parameters

3.4.1 Bluetooth scan switch (040)

To inquire and setup BLE scan switch.

Flag	Msg id offset	Parameter name	Parameter value
Write/read	40	scan_switch	number. 0-1 0: off 1: on

Inquire:

```
{
    "msg_id":  2040,
    "device_info":  {
        "mac":  "4c11ae8be624"
    }
}
```

Device reply:

```
{
    "msg_id": 2040,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "scan_switch": 0
    }
}
```

Setup:

```
{
    "msg_id": 1040,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "scan_switch": 0
    }
}
```

Device reply:

```
{
    "msg_id": 1040,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "result_code": 0,
    "result_msg": "success"
}
```

3.4.2 Filter relation (041)

To inquire and setup the relation of the MAC filter (043), adv name filter (044) and raw data filter (045). If the filters have been setup, but the relation does not include the filter, the filter will not take effect.

Flag	Msg id offset	Parameter name	Parameter value
Write/read	41	relation	number. 0-7 0: none 1: mac 2: adv name 3: raw data 4: adv name& raw data 5: mac& adv name& raw

			data 6: adv name| raw data 7: ADV name& mac **8: ADV NAME / MAC / Raw data (Added in V2.X.X firmware)**

Inquire:

```
{
    "msg_id": 2041,
    "device_info": {
        "mac": "4c11ae8be624"
    }
}
```

Device reply:

```
{
    "msg_id": 2041,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "relation": 4
    }
}
```

Setup:

```
{
    "msg_id": 1041,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "relation": 4
    }
}
```

Device reply:

```
{
    "msg_id": 1041,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "result_code": 0,
    "result_msg": "success"
```

```
}
```

3.4.3 RSSI filter (042)

To inquire and setup the Bluetooth signal strength filter.

Flag	Msg id offset	Parameter name	Parameter value
Write/read	42	rssi	number. -127~0, dBm

Inquire:

```
{
    "msg_id": 2042,
    "device_info": {
        "mac": "4c11ae8be624"
    }
}
```

Device reply:

```
{
    "msg_id": 2042,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "rssi": -127
    }
}
```

Setup:

```
{
    "msg_id": 1042,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "rssi": -127
    }
}
```

Device reply:

```
{
    "msg_id": 1042,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "result_code": 0,
    "result_msg": "success"
```

```
}
```

3.4.4 Mac address filter (043)

To inquire and setup the BLE MAC address filter.

Flag	Msg id offset	Parameter name	Parameter value
Write/read	43	precise	number. 0-1 0: Contains match (The MAC address contains the filter values) 1: Prefix match (The MAC address first N bytes must be the same as filter values)
		reverse	number. 0-1 0: positive filter (whitelist) 1: reverse filter (blacklist)
		mac	array. 0-10 group mac address, 0 means no filter. Each mac address is 2-12 characters (1-6 bytes).

Inquire:

```
{
    "msg_id": 2043,
    "device_info": {
        "mac": "4c11ae8be624"
    }
}
```

Device reply:

```
{
    "msg_id": 2043,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "precise": 0,
        "reverse": 0,
        "mac": [ ]
    }
}
```

Setup:

```
{
```

```
    "msg_id": 1043,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "precise": 0,
        "reverse": 0,
        "mac": [
            "112233",
            "445566"
        ]
    }
}
```

Device reply:

```
{
    "msg_id": 1043,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "result_code": 0,
    "result_msg": "success"
}
```

3.4.5 ADV name filter (044)

To inquire and setup the BLE advertising name filter.

Flag	Msg id offset	Parameter name	Parameter value
Write/read	44	precise	number. 0-1 0: Contains match (The ADV name contains the filter values) 1: Prefix match (The ADV name first N bytes must be the same as filter values)
		reverse	number. 0-1 0: positive filter (whitelist) 1: reverse filter (blacklist)
		name	array, 0-10 groups adv name, 0 means no filter. Each adv name is 1-20 characters.

Inquire:

```
{
```

```
    "msg_id":  2044,
    "device_info":  {
        "mac":  "4c11ae8be624"
    }
}
```

Device reply:

```
{
    "msg_id":  2044,
    "device_info":  {
        "mac":  "4c11ae8be624"
    },
    "data":  {
        "precise":  0,
        "reverse":  0,
        "name":  [ ]
    }
}
```

Setup:

```
{
    "msg_id":  1044,
    "device_info":  {
        "mac":  "4c11ae8be624"
    },
    "data":  {
        "precise":  0,
        "reverse":  0,
        "name":  [
            "MK110",
            "MK107"
        ]
    }
}
```

Device reply:

```
{
    "msg_id":  1044,
    "device_info":  {
        "mac":  "4c11ae8be624"
    },
    "result_code":  0,
    "result_msg":  "success"
}
```

3.4.6 Raw data filter switch (045)

To inquire the filter switch of BLE raw data types.

Flag	Msg id offset	Parameter name	Parameter value
Write/Read	45	ibeacon	number. 0: off, 1: on
		eddystone_uid	number. 0: off, 1: on
		eddystone_url	number. 0: off, 1: on
		eddystone_tlm	number. 0: off, 1: on
		bxp_devinfo	number. 0: off, 1: on
		bxp_acc	number. 0: off, 1: on
		bxp_th	number. 0: off, 1: on
		bxp_button	number. 0: off, 1: on
		bxp_tag	number. 0: off, 1: on
		pir	number. 0: off, 1: on
		other	number. 0: off, 1: on
		mk_tof	number. 0: off, 1: on
		nano_beacon_info	**number.** **0: off, 1: on** **Added in firmware V2.x.x**

Inquire:

```
{
    "msg_id": 2045,
    "device_info": {
        "mac": "4c11ae8be624"
    }
}
```

Device reply:

```
{
    "msg_id": 2045,
    "device_info": {
```

```
            "mac":  "4c11ae8be624"
      },
      "data":  {
            "ibeacon":  1,
            "eddystone_uid":  1,
            "eddystone_url":  1,
            "eddystone_tlm":  1,
            "bxp_devinfo":  1,
            "bxp_acc":  1,
            "bxp_th":  1,
            "bxp_button":  1,
            "bxp_tag":  1,
            "pir":  1,
            "other":  1,
            "mk_tof":  1,
            "nano_beacon_info":  1
      }
}
```

Setup:

```
{
      "msg_id":  1045,
      "device_info":  {
            "mac":  "4c11ae8be624"
      },
     "data":  {
            "ibeacon":  1,
            "eddystone_uid":  1,
            "eddystone_url":  1,
            "eddystone_tlm":  1,
            "bxp_devinfo":  1,
            "bxp_acc":  1,
            "bxp_th":  1,
            "bxp_button":  1,
            "bxp_tag":  1,
            "pir":  1,
            "other":  1,
            "mk_tof":  1,
            "nano_beacon_info":  1
      }

}
```

Device reply:

```
{
```

```
    "msg_id": 1045,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "result_code": 0,
    "result_msg": "success"
}
```

3.4.7 iBeacon filter (046)

To inquire and set up the filter rule for iBeacon frames.

Flag	Msg id offset	Parameter name	Parameter value
Write/read	46	switch_value	number. 0: off (rejected) 1: on (allowed)
		min_major	number. 0-65535
		max_major	number. 0-65535 max major>= min major
		min_minor	number. 0-65535
		max_minor	number. 0-65535 max minor>= min minor
		uuid	string. 0-32 characters (0-16 bytes). If set it empty, it will allow all UUIDs reported.

Inquire:

```
{
    "msg_id": 2046,
    "device_info": {
        "mac": "4c11ae8be624"
    }
}
```

Device reply:

```
{
    "msg_id": 2046,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "switch_value": 1,
        "min_major": 0,
        "max_major": 65535,
```

```
            "min_minor": 0,
            "max_minor": 65535,
            "uuid": ""
        }
}
```

Setup:

```
{
    "msg_id": 1046,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "switch_value": 1,
        "min_major": 0,
        "max_major": 65535,
        "min_minor": 0,
        "max_minor": 65535,
        "uuid": ""
    }
}
```

Device reply:

```
{
    "msg_id": 1046,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "result_code": 0,
    "result_msg": "success"
}
```

3.4.8 Eddystone-UID filter (047)

To inquire and set up the filter rule for Eddystone-UID frame.

Flag	Msg id offset	Parameter name	Parameter value
Write/read	47	switch_value	number. 0: off (rejected) 1: on (allowed)
		namespace	string. 0-20 characters (0-10 bytes) If set it empty, it will allow all namespace reported.
		instance	string.

			0-12 characters (0-6 bytes) If set it empty, it will allow all instance reported.

Inquire:

```
{
    "msg_id": 2047,
    "device_info": {
        "mac": "4c11ae8be624"
    }
}
```

Device reply:

```
{
    "msg_id": 2047,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "switch_value": 1,
        "namespace": "",
        "instance": ""
    }
}
```

Setup:

```
{
    "msg_id": 1047,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "switch_value": 1,
        "namespace": "",
        "instance": ""
    }
}
```

Device reply:

```
{
    "msg_id": 1047,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "result_code": 0,
    "result_msg": "success"
```

```
}
```

3.4.9 Eddystone-URL filter (048)

To inquire and set up the filter rule for Eddystone-URL frame.

Flag	Msg id offset	Parameter name	Parameter value
Write/read	48	switch_value	number. 0: off (rejected) 1: on (allowed)
		url	string. 0-37 characters If set it empty, it will allow all url reported.

Inquire:

```
{
    "msg_id": 2048,
    "device_info": {
        "mac": "4c11ae8be624"
    }
}
```

Device reply:

```
{
    "msg_id": 2048,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "switch_value": 1,
        "url": ""
    }
}
```

Setup:

```
{
    "msg_id": 1048,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "switch_value": 1,
        "url": "https://mokosmart.com"
    }
}
```

Device reply:

```
{
    "msg_id": 1048,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "result_code": 0,
    "result_msg": "success"
}
```

3.4.10 Eddystone-TLM filter (049)

To inquire and set up the filter rule for Eddystone-TLM frame.

Flag	Msg id offset	Parameter name	Parameter value
Write/read	49	switch_value	number. 0: off (rejected) 1: on (allowed)
		tlm_version	number. 0-2 0: version 0 1: version 1 2: all versions

Inquire:

```
{
    "msg_id": 2049,
    "device_info": {
        "mac": "4c11ae8be624"
    }
}
```

Device reply:

```
{
    "msg_id": 2049,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "switch_value": 1,
        "tlm_version": 0
    }
}
```

Setup:

```
{
    "msg_id": 1049,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "switch_value": 1,
        "tlm_version": 0
    }
}
```

Device reply:

```
{
    "msg_id": 1049,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "result_code": 0,
    "result_msg": "success"
}
```

3.4.11 BXP- device info filter (050)

To inquire and set up the filter rule for MOKO BXP-device info frames.

Flag	Msg id offset	Parameter name	Parameter value
Write/read	50	switch_value	number. 0: off (rejected) 1: on (allowed)

Inquire:

```
{
    "msg_id": 2050,
    "device_info": {
        "mac": "4c11ae8be624"
    }
}
```

Device reply:

```
{
    "msg_id": 2050,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
```

```
            "switch_value":  1
        }
}
```

Setup:

```
{
        "msg_id":  1050,
        "device_info":  {
            "mac":  "4c11ae8be624"
        },
        "data":  {
            "switch_value":  1
        }
}
```

Device reply:

```
{
        "msg_id":  1050,
        "device_info":  {
            "mac":  "4c11ae8be624"
        },
        "result_code":  0,
        "result_msg":  "success"
}
```

3.4.12 BXP-ACC filter (051)

To inquire and set up the filter rule for MOKO BXP-ACC frames.

Flag	Msg id offset	Parameter name	Parameter value
Write/read	51	switch_value	number. 0: off (rejected) 1: on (allowed)

Inquire:

```
{
        "msg_id":  2051,
        "device_info":  {
            "mac":  "4c11ae8be624"
        }
}
```

Device reply:

```
{
        "msg_id":  2051,
        "device_info":  {
```

```
        "mac": "4c11ae8be624"
    },
    "data": {
        "switch_value": 1
    }
}
```

Setup:

```
{
    "msg_id": 1051,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "switch_value": 1
    }
}
```

Device reply:

```
{
    "msg_id": 1051,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "result_code": 0,
    "result_msg": "success"
}
```

3.4.13 BXP-TH filter (052)

To inquire and set up the filter rule for MOKO BXP-TH frames.

Flag	Msg id offset	Parameter name	Parameter value
Write/read	52	switch_value	number. 0: off (rejected) 1: on (allowed)

Inquire:

```
{
    "msg_id": 2052,
    "device_info": {
        "mac": "4c11ae8be624"
    }
}
```

Device reply:

```
{
    "msg_id": 2052,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "switch_value": 1
    }
}
```

Setup:

```
{
    "msg_id": 1052,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "switch_value": 1
    }
}
```

Device reply:

```
{
    "msg_id": 1052,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "result_code": 0,
    "result_msg": "success"
}
```

3.4.14 BXP-Button filter (053)

To inquire and set up the filter rule for MOKO BXP-Button frames.

Flag	Msg id offset	Parameter name	Parameter value
Write/read	53	switch_value	number. 0: off (rejected) 1: on (allowed)
		single_press	Single press trigger frame number. 0: off (rejected) 1: on (allowed)
		double_press	Double press trigger

			frame number. 0: off (rejected) 1: on (allowed)
		long_press	Long press trigger frame. number. 0: off (rejected) 1: on (allowed)
		abnormal_inactivity	Abnormal inactivity trigger frame number. 0: off (rejected) 1: on (allowed)

Inquire:

```
{
    "msg_id": 2053,
    "device_info": {
        "mac": "4c11ae8be624"
    }
}
```

Device reply:

```
{
    "msg_id": 2053,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "switch_value": 1,
        "single_press": 1,
        "double_press": 1,
        "long_press": 1,
        "abnormal_inactivity": 1
    }
}
```

Setup:

```
{
    "msg_id": 1053,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "switch_value": 1,
```

```
            "single_press":  1,
            "double_press":  1,
            "long_press":  1,
            "abnormal_inactivity":  1
        }
}
```

Device reply:

```
{
    "msg_id":  1053,
    "device_info":  {
            "mac":  "4c11ae8be624"
    },
    "result_code":  0,
    "result_msg":  "success"
}
```

3.4.15 BXP-Tag filter (054)

To inquire and set up the filter rule for MOKO BXP-Tag frames.

Flag	Msg id offset	Parameter name	Parameter value
Write/read	54	switch_value	number. 0: off (rejected) 1: on (allowed)
		precise	number. 0-1 0: Contains match (The tag id contains the filter values) 1: Prefix match (The tag id first N bytes must be the same as filter values)
		reverse	number. 0-1 0: positive filter (whitelist) 1: reverse filter (blacklist)
		tagid	array. 0-10 groups tag id. Each tag id is 2-12 characters (1-6 bytes).

Inquire:

```
{
    "msg_id":  2054,
    "device_info":  {
            "mac":  "4c11ae8be624"
    }
}
```

Device reply:

```
{
    "msg_id": 2054,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "switch_value": 1,
        "precise": 0,
        "reverse": 0,
        "tagid": [ ]
    }
}
```

Setup:

```
{
    "msg_id": 1054,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "switch_value": 1,
        "precise": 0,
        "reverse": 0,
        "tagid": [
            "000",
            "111"
            ]
    }
}
```

Device reply:

```
{
    "msg_id": 1054,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "result_code": 0,
    "result_msg": "success"
}
```

3.4.16 MK PIR filter (055)

To inquire and set up the filter rule for MOKO PIR frames.

Flag	Msg id offset	Parameter name	Parameter value
Write/read	55	switch_value	number. 0: off (rejected) 1: on (allowed)
		delay_response_status	number. 0-3 0: low delay 1: medium delay 2: high delay 3: all types
		door_status	number. 0-2 0: close 1: open 2: all types
		sensor_sensitivity	number. 0-3 0: low sensitivity 1: medium sensitivity 2: high sensitivity 3: all types
		sensor_detection_status	number. 0-2 0: no motion detected 1: motion detected 2: all status
		min_major	number. 0-65535
		max_major	number. 0-65535 max major>= min major
		min_minor	number. 0-65535
		max_minor	number. 0-65535 max minor>= min minor

Inquire:

```
{
    "msg_id": 2055,
    "device_info": {
        "mac": "4c11ae8be624"
    }
}
```

Device reply:

```
{
    "msg_id": 2055,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
```

```
            "switch_value": 1,
            "delay_response_status": 3,
            "door_status": 2,
            "sensor_sensitivity": 3,
            "sensor_detection_status": 2,
            "min_major": 0,
            "max_major": 0,
            "min_minor": 0,
            "max_minor": 0
        }
}
```

Setup:

```
{
    "msg_id": 1055,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "switch_value": 1,
        "delay_response_status": 3,
        "door_status": 2,
        "sensor_sensitivity": 3,
        "sensor_detection_status": 2,
        "min_major": 0,
        "max_major": 0,
        "min_minor": 0,
        "max_minor": 0
    }
}
```

Device reply:

```
{
    "msg_id": 1055,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "result_code": 0,
    "result_msg": "success"
}
```

3.4.17 Other type filter (056)

To inquire and set up the filter rule for other type frames.

Flag	Msg id offset	Parameter name	Parameter value
Write/read	56	switch_value	number. 0: off (rejected) 1: on (allowed)
		relation	number. 0-5 0: A 1: A&B 2: A|B 3: A&B&C 4: (A&B)|C 5: A|B|C
		rule	array. Can set at most 3 groups filter rules.
		type	Data of the array. String, 2 characters (1 byte). BLE Data type, 00 means do not filter by data type.
		start	Data of the array. number: 0-29. The start byte under the data type.
		end	Data of the array. number: 0-29. The end byte under the data type, should be greater than the start byte.
		raw_data	Data of the array, string. The BLE raw data from the start to end byte under the data type.

Inquire:

```
{
    "msg_id": 2056,
    "device_info": {
        "mac": "4c11ae8be624"
    }
}
```

Device reply：

```
{
    "msg_id": 2056,
    "device_info": {
        "mac": "4c11ae8be624"
```

```
    },
    "data": {
        "switch_value": 1,
        "relation": 5,
        "rule": [ ]
    }
}
```

Setup：

```
{
    "msg_id": 1056,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "switch_value": 1,
        "relation": 0,
        "rule": [
            {
                "type": "FF",
                "start": 1,
                "end": 2,
                "raw_data": "004C"
            }
        ]
    }
}
```

Device reply:

```
{
    "msg_id": 1056,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "result_code": 0,
    "result_msg": "success"
}
```

3.4.18 Duplicate data filter (057)

To inquire and set up the rule to filter out duplicate advertisements.

Flag	Msg id offset	Parameter name	Parameter value
Write/read	57	~~timeout~~	~~number.~~ ~~Filter period: 1-86400, unit:s~~

			Deleted in firmware V2.x.x
		rule	number. 0-3 0: disable 1: mac 2: mac and data type 3: mac and raw data

Inquire:

```
{
    "msg_id": 2057,
    "device_info": {
        "mac": "4c11ae8be624"
    }
}
```

Device reply:

```
{
    "msg_id": 2057,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "rule": 1
    }
}
```

Setup:

```
{
    "msg_id": 1057,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "rule": 1
    }
}
```

Device reply:

```
{
    "msg_id": 1057,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "result_code": 0,
    "result_msg": "success"
```

```
}
```

3.4.19 Bluetooth data report setting (059)

To inquire and set the Bluetooth data report items.

Flag	Msg id offset	Parameter name	Parameter value
Write/read	59	timestamp	number. 0-1 0: not upload, 1: upload
		adv_data	advertisement data. 0: not upload 1: upload
		~~rsp_data~~	~~Scan response data~~ ~~0: not upload~~ ~~1: upload~~ **Deleted in Firmware V2.x.x**
		~~parse_adv_data~~	~~number. 0-1~~ ~~0: not upload parsed data~~ ~~1: upload parsed data~~ **Deleted in firmware V2.x.x**

Inquire:

```
{
    "msg_id": 2059,
    "device_info": {
        "mac": "4c11ae8be624"
    }
}
```

Device reply:

```
{
    "msg_id": 2059,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "timestamp": 1,
        "adv_data": 1,
    }
}
```

Setup:

```
{
    "msg_id": 1059,
```

```
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "timestamp": 1,
        "adv_data": 1
    }
}
```

Device reply:

```
{
    "msg_id": 1059,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "result_code": 0,
    "result_msg": "success"
}
```

3.4.20 PHY filter (060)

To inquire and setup the PHY filter.

Flag	Msg id offset	Parameter name	Parameter value
Write/read	60	phy_filter	number. 0-3 0：1M PHY (V4.2) 1：1M PHY (V5.0) 2：1M PHY (V4.2) & 1M PHY (V5.0) 3：Coded PHY (V5.0) 4：1M PHY(V4.2) & 1M PHY(V5.0)& Coded **(Added in V2.X.X firmware)**

Inquire:

```
{
    "msg_id": 2060,
    "device_info": {
        "mac": "4c11ae8be624"
    }
}
```

Device reply:

```
{
    "msg_id": 2060,
    "device_info": {
```

```
        "mac": "4c11ae8be624"
    },
    "data": {
        "phy_filter": 1
    }
}
```

Setup:

```
{
    "msg_id": 1060,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "phy_filter": 1
    }
}
```

Device reply:

```
{
    "msg_id": 1060,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "result_code": 0,
    "result_msg": "success"
}
```

3.4.21 Gateway advertisement settings(061)

The gateway can work as a scanner and advertiser simultaneously. The gateway advertisement frame conforms standard iBeacon.

This command is used to inquire and setup the iBeacon advertising parameters.

Flag	Msg id offset	Parameter name	Parameter value
Write/read	61	switch_value	number. 0-1 0: off, 1: on
		major	number. 0-65535
		minor	number. 0-65535
		uuid	string. 32 characters (16 bytes)
		rssi_1m	number. -100~0
		adv_interval	number. 1-100, unit: 100ms
		tx_power	number. 0-15 0: -24 dBm

			1: -21 dBm 2: -18 dBm 3: -15 dBm 4: -12 dBm 5: -9 dBm 6: -6 dBm 7: -3 dBm 8: 0 dBm 9: 3 dBm 10: 6 dBm 11: 9 dBm 12: 12 dBm 13: 15 dBm 14: 18 dBm 15: 21 dBm
		connectable	number. 0/1

Inquire:

```
{
    "msg_id": 2061,
    "device_info": {
        "mac": "4c11ae8be624"
    }
}
```

Device reply:

```
{
    "msg_id": 2061,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "switch_value": 1,
        "major": 100,
        "minor": 99,
        "uuid": "00112233445566778899001122334455",
        "rssi_1m": -59,
        "adv_interval": 1,
        "tx_power": 9
        "connectable": 1
    }
}
```

Setup:

```
{
```

```
    "msg_id": 1061,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "switch_value": 1,
        "major": 100,
        "minor": 99,
        "uuid": "00112233445566778899001122334455",
        "rssi_1m": -59,
        "adv_interval": 1,
        "tx_power": 9
        "connectable": 1

    }
}
```

Device reply:

```
{
    "msg_id": 1061,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "result_code": 0,
    "result_msg": "success"
}
```

3.4.22 MK TOF Filter (062)

To inquire and setup the tof filter.

Flag	Msg id offset	Parameter name	Parameter value
Write/read	62	switch_value	number. 0: off (rejected) 1: on (allowed)
		mfg_code	Array, 0-10 groups mfg codes. Each mfg code is 4 characters (2 bytes).

Inquire:

```
{
    "msg_id": 2062,
    "device_info": {
        "mac": "4c11ae8be624"
    }
```

```
}
```

Device reply:

```
{
    "msg_id": 2062,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "switch_value": 1,
        "mfg_code": [ ]
    }
}
```

Setup:

```
{
    "msg_id": 1062,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
         "switch_value": 1,
        "mfg_code": ["0059"]

    }
}
```

Device reply:

```
{
    "msg_id": 1062,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "result_code": 0,
    "result_msg": "success"
}
```

3.4.23 Bluetooth data report interval (063)

To inquire and setup the Bluetooth data report interval. If set the interval to 0, the gateway will perform real-time reporting.

This command is supported on Firmware V2.X.X, not supported in firmware V1.X.X.

Flag	Msg id offset	Parameter name	Parameter value
Write/read	63	interval	number. 0-86400, unit: s default: 0

Inquire:

```
{
    "msg_id": 2063,
    "device_info": {
        "mac": "4c11ae8be624"
    }
}
```

Device reply:

```
{
    "msg_id": 2063,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "interval": 600
    }
}
```

Setup:

```
{
    "msg_id": 1063,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "interval": 600
    }
}
```

Device reply:

```
{
    "msg_id": 1063,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "result_code": 0,
    "result_msg": "success"
}
```

3.4.24 NanoBeacon filter (064)

To inquire and set up the filter rule for MOKO NanoBeacon info frame.

This command is supported on Firmware V2.X.X, not supported in firmware V1.X.X.

Flag	Msg id offset	Parameter name	Parameter value
Write/read	64	switch_value	number. 0: off (rejected) 1: on (allowed)

Inquire:

```
{
    "msg_id": 2064,
    "device_info": {
        "mac": "4c11ae8be624"
    }
}
```

Device reply:

```
{
    "msg_id": 2064,
    "device_info": {
        "mac": "4c11ae8be624"
    },
   "data": {
        "switch_value": 1
    }
}
```

Setup:

```
{
    "msg_id": 1064,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
         "switch_value": 1
    }
}
```

Device reply:

```
{
    "msg_id": 1064,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "result_code": 0,
    "result_msg": "success"
}
```

3.4.25 Data parsing Setting (065)

To inquire and set up the Bluetooth data parsing.

This command is supported on Firmware V2.X.X, not supported in firmware V1.X.X.

Flag	Msg id offset	Parameter name	Parameter value
Write/read	65	iBeacon_switch	number. 0: off (not parse) 1: on (parse)
		eddystone_uid_switch	number. 0: off (not parse) 1: on (parse)
		eddystone_tlm_switch	number. 0: off (not parse) 1: on (parse)
		eddystone_url_switch	number. 0: off (not parse) 1: on (parse)
		bxp_acc_switch	number. 0: off (not parse) 1: on (parse)
		bxp_th_switch	number. 0: off (not parse) 1: on (parse)
		bxp_devinfo_switch	number. 0: off (not parse) 1: on (parse)
		bxp_button_switch	number. 0: off (not parse) 1: on (parse)
		bxp_tag&sensor_switch	number. 0: off (not parse) 1: on (parse)
		mk_pir_switch	number. 0: off (not parse) 1: on (parse)
		mk_tof_switch	number. 0: off (not parse) 1: on (parse)
		nanobeacon_info_switch	number. 0: off (not parse) 1: on (parse)

Inquire:

```
{
      "msg_id":  2065,
      "device_info":  {
            "mac":  "4c11ae8be624"
      }
}
```

Device reply:

```
{
   "msg_id":  2065,
   "device_info":  {
      "mac":  "4c11ae8be624"
   },
   "data":  {
      "iBeacon_switch":  1,
      "eddystone_uid_switch":  1,
      "eddystone_tlm_switch":  1,
      "eddystone_url_switch":  1,
      "bxp_acc_switch":  1,
      "bxp_th_switch":  1,
      "bxp_devinfo_switch":  1,
      "bxp_button_switch":  1,
      "bxp_tag&sensor_switch":  1,
      "mk_pir_switch":  1,
      "mk_tof_switch":  1,
      "nanobeacon_info_switch":  1
   }
}
```

Setup:

```
{
      "msg_id":  1065,
      "device_info":  {
            "mac":  "4c11ae8be624"
      },
      "data":  {
      "iBeacon_switch":  1,
      "eddystone_uid_switch":  1,
      "eddystone_tlm_switch":  1,
      "eddystone_url_switch":  1,
      "bxp_acc_switch":  1,
      "bxp_th_switch":  1,
      "bxp_devinfo_switch":  1,
      "bxp_button_switch":  1,
      "bxp_tag&sensor_switch":  1,
```

```
    "mk_pir_switch": 1,
    "mk_tof_switch": 1,
    "nanobeacon_info_switch": 1
  }
}
```

Device reply:

```
{
    "msg_id": 1065,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "result_code": 0,
    "result_msg": "success"
}
```

3.4.26 Bluetooth Scan Mode(066)

To inquire and setup the Bluetooth scan mode.

This command is supported on Firmware V2.X.X, not supported in firmware V1.X.X.

Flag	Msg id offset	Parameter name	Parameter value
Write/read	66	scan_mode	number. 0: Active scan 1: Passive scan

Inquire:

```
{
    "msg_id": 2066,
    "device_info": {
        "mac": "4c11ae8be624"
    }
}
```

Device reply:

```
{
    "msg_id": 2066,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "scan_mode": 0
    }
}
```

Setup:

```
{
	"msg_id":  1066,
	"device_info":  {
		"mac":  "4c11ae8be624"
	},
	"data":  {
		"scan_mode":  1
	}
}
```

Device reply:

```
{
	"msg_id":  1066,
	"device_info":  {
		"mac":  "4c11ae8be624"
	},
	"result_code":  0,
	"result_msg":  "success"
}
```

3.5 Report Bluetooth data (070)

When the gateway scans the Beacon data, it will combine advertisements to packet and report to cloud.

The gateway will automatically parse the iBeacon, Eddystone and MOKO beacon frames, then upload the parsed data to cloud. While for other type frames, the gateway will upload their raw data directly.

Flag	Msg id offset	Parameter name	Parameter value
Notify	70	timestamp	Unix timestamp, millisecond level.
		timezone	Step by half timezone.
		type_code	number. 0-12 0: ibeacon 1: eddystone-uid 2: eddystone-url 3: eddystone-tlm 4: bxp-devinfo 5: bxp-acc 6: bxp-th 7: bxp-button 8: bxp-tag 9: pir 10: other 11: mk tof 12:nano beacon info **11 and 12 were added in firmware V2.X.X**
		type	The description for type code
		rssi	number.
		connectable	number. 0-1 0: not connectable 1: connectable
		mac	mac address of Beacon
		raw_data	Advertising raw data.
		Varies contents	Parsed data, content varies in different frames. See the following chapters.

3.5.1 iBeacon frame


```
{
  "msg_id" : 3070,
  "device_info" : {
    "mac" : "083a8d8bc9a0"
  },
  "data" : [ {
    "timestamp" : 1686738414798,
    "timezone" : -24,
    "type_code" : 0,
    "type" : "ibeacon",
    "rssi" : -93,
    "connectable" : 0,
    "mac" : "665544332202",
    "uuid" : "e2c56db5dffb48d2b060d0f5a71096e0",
    "major" : 1,
    "minor" : 2,
    "rssi_1m" : -56
  } ]
}
```


3.5.2 Eddystone-UID frame


```
{
  "msg_id" : 3070,
  "device_info" : {
    "mac" : "083a8d8bc9a0"
  },
  "data" : [ {
    "timestamp" : 1686741768642,
    "timezone" : -24,
    "type_code" : 1,
    "type" : "eddystone-uid",
    "rssi" : -81,
    "connectable" : 1,
    "mac" : "e942f7285a79",
    "namespace" : "adb143488994071c8d43",
    "instance" : "a00000000001",
    "rssi_0m" : 0
  } ]
}
```


3.5.3 Eddystone-URL frame


```
{
  "msg_id" : 3070,
  "device_info" : {
    "mac" : "083a8d8bc9a0"
  },
  "data" : [ {
    "timestamp" : 1686741869955,
    "timezone" : -24,
    "type_code" : 2,
    "type" : "eddystone-url",
    "rssi" : -93,
    "connectable" : 0,
    "mac" : "ef5aaac187c0",
    "url" : "https://www.mokosmart.com",
    "rssi_0m" : 0
  } ]
}
```

3.5.4 Eddystone-TLM frame


```
{
  "msg_id" : 3070,
  "device_info" : {
    "mac" : "083a8d8bc9a0"
  },
  "data" : [ {
    "timestamp" : 1686741926655,
    "timezone" : -24,
    "type_code" : 3,
    "type" : "eddystone-tlm",
    "rssi" : -75,
    "connectable" : 1,
    "mac" : "e0798d04a768",
    "tlm_version" : 0,
    "batt_vol" : 3023,
    "temperature" : 27.3,
    "adv_count" : 577,
    "runtime" : "17d9h49m34.2S"
  } ]
}
```


3.5.5 BXP-device info frame

Add sensor status in V2.X.X firmware:


```
{
  "msg_id" : 3070,
  "device_info" : {
    "mac" : "083a8d8bc9a0"
  },
  "data" : [ {
    "timestamp" : 1686741961892,
    "timezone" : -24,
    "type_code" : 4,
    "type" : "bxp-devinfo",
    "rssi" : -75,
    "connectable" : 1,
    "mac" : "f0c9b61e2854",
    "ranging_data" : -65,
    "adv_interval" : 1000,
    "batt_vol" : 2914,
    "passwd_verification" : 1,
    "have_light_sensor" : 0,
    "light_sensor_status" : 0,
    "firmware_ver" : "V3.3.7"
  } ]
}
```


V1.X.X firmware


```
{
  "msg_id" : 3070,
  "device_info" : {
    "mac" : "142b2fe9b79c"
  },
  "data" : [ {
    "timestamp" : 1780886967692,
    "timezone" : 0,
    "type_code" : 4,
    "type" : "bxp-devinfo",
    "rssi" : -99,
    "connectable" : 1,
    "mac" : "d3d6d04e9460",
    "ranging_data" : 0,
    "adv_interval" : 500,
    "batt_vol" : 3582,
    "passwd_verification" : 1,
    "have_light_sensor" : 0,
    "have_hall_sensor" : 0,
    "have_tamper_sensor" : 0,
    "light_sensor_status" : 0,
    "door_status" : 0,
    "tamper_status" : 0,
    "firmware_ver" : "V3.0.6"
  } ]
}
```


V2.X.X firmware

3.5.6 BXP-ACC frame


```
{
  "msg_id" : 3070,
  "device_info" : {
    "mac" : "083a8d8bc9a0"
  },
  "data" : [ {
    "timestamp" : 1686742061404,
    "timezone" : -24,
    "type_code" : 5,
    "type" : "bxp-acc",
    "rssi" : -82,
    "connectable" : 1,
    "mac" : "e23e4e4ae6b3",
    "ranging_data" : 2,
    "adv_interval" : 1000,
    "sampling_rate" : 10,
    "full_scale" : "2g",
    "motion_threshold" : 100,
    "x_axis_data" : 64,
    "y_axis_data" : -5056,
    "z_axis_data" : 15488,
    "batt_vol" : 3024
  } ]
}
```


3.5.7 BXP-TH frame

Add device type in V2.X.X firmware:


```
{
  "msg_id" : 3070,
  "device_info" : {
    "mac" : "083a8d8bc9a0"
  },
  "data" : [ {
    "timestamp" : 1686742096971,
    "timezone" : -24,
    "type_code" : 6,
    "type" : "bxp-th",
    "rssi" : -84,
    "connectable" : 1,
    "mac" : "c3d7c95e2813",
    "txpower" : 0,
    "ranging_data" : 0,
    "adv_interval" : 500,
    "temperature" : 26.8,
    "humidity" : 56.4,
    "batt_vol" : 3447
  } ]
}
```


V1.X.X firmware


```
{
  "msg_id" : 3070,
  "device_info" : {
    "mac" : "142b2fe9b79c"
  },
  "data" : [ {
    "timestamp" : 1780887261308,
    "timezone" : 0,
    "type_code" : 6,
    "type" : "bxp-th",
    "rssi" : -97,
    "connectable" : 1,
    "mac" : "e8f3367e0f39",
    "txpower" : 0,
    "ranging_data" : 0,
    "adv_interval" : 1000,
    "temperature" : -21.1,
    "humidity" : 77.9,
    "batt_vol" : 3392,
    "device_type" : 3
  } ]
}
```

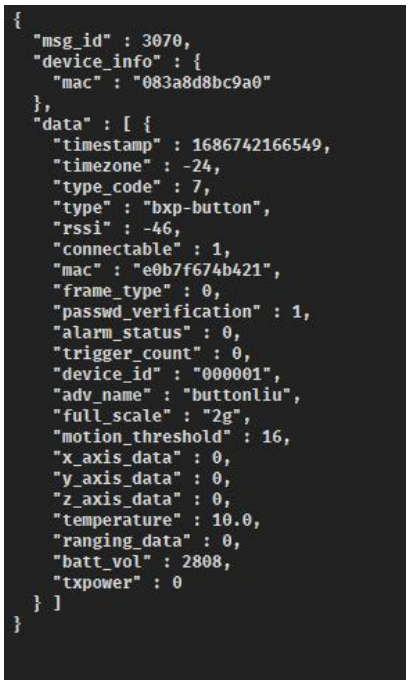

V2.X.X firmware

3.5.8 BXP-Button frame

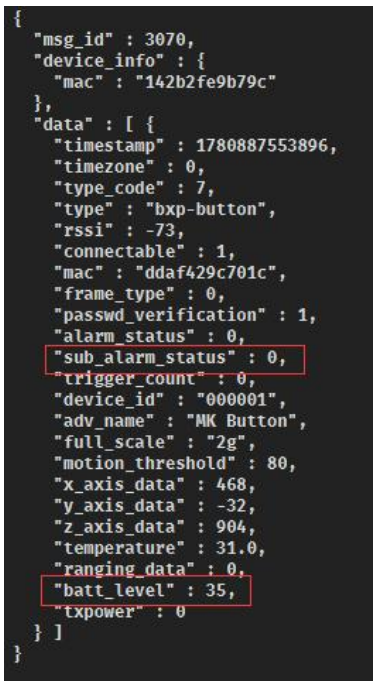
Changes in V2.X.X firmware:

Added sub button alarm status;

Distinguished battery data: If the advertisement contains battery voltage, gateway will report batt_vol, if the advertisement contains battery percent, gateway will report batt_level.



V1.X.X firmware



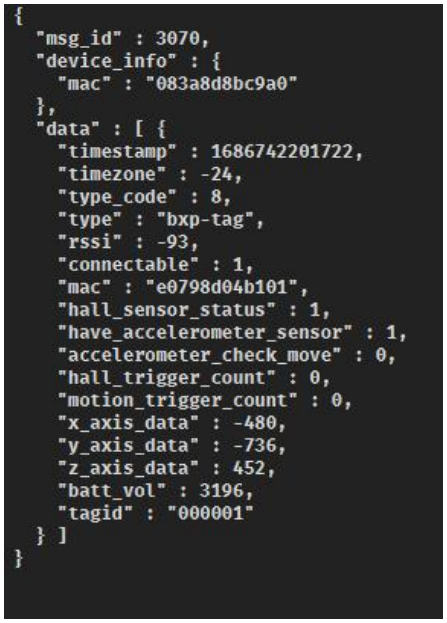
V2.X.X firmware

3.5.9 BXP-Tag & Sensor frame

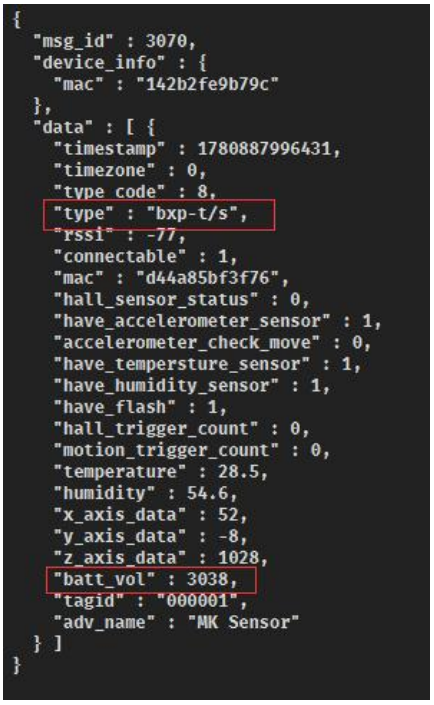
Changes in V2.X.X firmware:

Changed the type name;

Distinguished battery data: If the advertisement contains battery voltage, gateway will report batt_vol, if the advertisement contains battery percent, gateway will report batt_level.



V1.X.X firmware

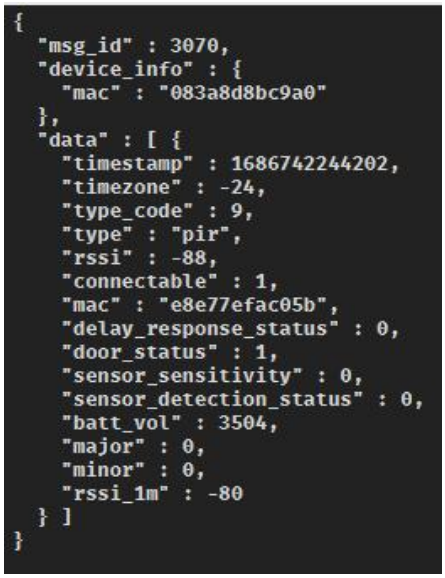


V2.X.X firmware

3.5.10 MK PIR frame

Changes in V2.X.X firmware:

Added adv timestamp.



V1.X.X firmware



V2.X.X firmware

3.5.11 MK TOF frame


```
{
  "msg_id" : 3070,
  "device_info" : {
    "mac" : "142b2fe9b19c"
  },
  "data" : [ {
    "timestamp" : 1769161860114,
    "timezone" : 0,
    "type_code" : 11,
    "type" : "tof",
    "rssi" : -93,
    "connectable" : 1,
    "mac" : "cfb71ad15a3e",
    "raw_data" : "02010612ff5900076cfe0affffff1500ffff013e5aff",
    "beacon_code" : "6c07",
    "mfg_code" : "0059",
    "batt_vol" : 2814,
    "ranging_distance" : 21,
    "user_data" : "ffff",
    "sub_type" : 1
  } ]
}
```


3.5.12 NanoBeacon info frame


```
{
  "msg_id" : 3070,
  "device_info" : {
    "mac" : "142b2fe9b19c"
  },
  "data" : [ {
    "timestamp" : 1769161798378,
    "timezone" : 0,
    "type_code" : 12,
    "type" : "nano_beacon_info",
    "rssi" : -79,
    "connectable" : 0,
    "mac" : "ff2233445566",
    "raw_data" : "06094e4254414712ff050500669108d8840e00dd6655443322ff",
    "device_name" : "NBTAG",
    "mf_id" : "0505",
    "adv_type" : "normal",
    "batt_vol" : 3187,
    "temperature" : 21.9,
    "sec_cnt" : 951512,
    "trigger_status" : "no alarm"
  } ]
}
```

3.5.13 Other type frames


```
{
  "msg_id" : 3070,
  "device_info" : {
    "mac" : "142b2fe9b19c"
  },
  "data" : [ {
    "timestamp" : 1769162717181,
    "timezone" : 0,
    "type_code" : 10,
    "type" : "other",
    "rssi" : -47,
    "connectable" : 1,
    "mac" : "e80690605269",
    "raw_data" : "02010604160faa010b094d4b4757332d353236390fff0faae8069060526956322e302e35"
  } ]
}
```


4. Revision History

Revision	Change notes	Editor	Date
V1.0	Initial Version	Weiguifen	2023.10.10
V1.1	Improve description errors.	Weiguifen	2023.12.10
V1.2	Add chapter *3.3.6 Connect to MOKO H4 sensor*.	Weiguifen	2024.2.2
V1.3	Add chapter *3.3.5 Connect with MOKO Button*	Weiguifen	2024.8.22
V2.1	Applicable to V2 FW	Damon	2025.8.20

V2.2	Compatible with V2.X.X firmware. 1. Added parameters in *3.3.2 Device information* 2. Added parameters in *3.4.6 Raw data filter switch* 3. Deleted parameters in *3.4.18 Duplicate data filter* 4. Changed parameters in *3.4.19 Bluetooth data report setting* 5. Added command in *3.4.23 Bluetooth data report interval* and *3.4.24 NanoBeacon filter* 6. Added Beacon type in *3.5 Report Bluetooth data*	Weiguifen	2026.2.10
V2.3	1. Added result code in report WIFI firmware upgrade result and Bluetooth firmware upgrade result, *3.3.6 WIFI Firmware Upgrade Result & 3.3.14 Bluetooth Firmware Upgrade Result.* 2. Added value in *3.4.2 Filter Relation* 3. Deleted parameters in *3.4.19 Bluetooth data report setting* 4. Added value in *3.4.20 PHY Filter* 5. Added command in *3.4.25 Data Parse Setting* 6. Added command in *3.4.26 Bluetooth Scan Mode* 7. Added information in *3.5.5 BXP- Device info frame* 8. Added information in *3.5.7 BXP- TH frame* 9. Added information and distinguish battery data(battery voltage or percent) in *3.5.8 BXP- Button frame* 10. Changed type name and distinguish battery data(battery voltage or percent) in *3.5.9 BXP- Tag&Sensor frame* 11. Add adv timestamp in *3.5.10 MK PIR Frame* 12. Add NanoBeacon info frame in *3.5.12 NanoBeacon Info Frame*	Weiguifen	2026.6.5

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